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The role of hatchery origin in early marine residence and survival of Interior Columbia River spring and summer Chinook salmon (*Oncorhynchus tshawytscha*)

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Abstract

Interior Columbia River spring and summer (ICRSS) Chinook salmon are listed under the Endangered Species Act and experience persistently poor survival, despite low ocean intercept rates. Intra-annual variation in marine survival among hatcheries is often as large as interannual variation. Using parentage-based tagging, we assigned ICRSS yearlings to their hatchery of origin and evaluated how size and condition at capture varied across hatcheries and whether that variation was related to marine survival. We also compared timing and size at marine entry and early ocean growth across hatchery and wild origin yearlings. We observed consistent differences in early marine residence traits among hatcheries, indicating hatchery-specific influences on performance months after release. After accounting for interannual variability, hatchery of origin explained more of the variation (~47%) in marine survival than body length or condition at capture, hatchery distance from the ocean (~16%), or the number of dams passed during migration (~14%). These findings suggest that carry-over effects associated with hatchery of origin may play a substantial role in shaping early marine survival of ICRSS Chinook Salmon.

Keywords: Parentage-Based Tagging, Otoliths, Smolt-to-Adult Return, Carry-over Effects, Fisheries Management

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Introduction

Chinook salmon (*Oncorhynchus tshawytscha*) is a culturally and economically important species, supporting tribal and recreational fisheries throughout the Pacific Northwest. Historically, they were the most abundant salmon species in the Columbia River Basin, the second largest watershed in the United States (Fulton 1968). Over time, however, Chinook salmon populations have experienced substantial declines, driven in part by habitat fragmentation associated with the construction of large hydropower dams (Zabel and Williams 2002; Fig. 1). In response to these declines, Evolutionary Significant Units (ESU) of Interior Columbia River spring/summer (ICRSS) Chinook salmon were listed under the Endangered Species Act (ESA; Waples 1991). Among these, the Upper Columbia River spring run ESU is currently listed as endangered, and the Snake River spring/summer run ESU is listed as threatened. While the Mid-Columbia River spring run remains unlisted, it faces similar pressures (Northwest Fisheries Science Center 2015).

With their anadromous life cycle, Chinook salmon must navigate diverse habitats during their seaward migration, which collectively play a vital role in their overall survival and abundance (Magnusson and Hilborn 2003; Bond et al. 2008; Haeseker et al. 2012). Some smolts migrate from the freshwater to the ocean during their first year, referred to as subyearlings, while others complete a full year in the freshwater before migrating to sea, referred to as yearlings (Tucker et al. 2011). ICRSS Chinook salmon migrate as yearlings, and research indicates that freshwater survival rates during their out-migration to Bonneville Dam remain relatively stable with a mean of near 60% from 2015 to 2021 (Fish Passage Center 2025). However, estuarine and marine survival are considerably lower, often less than 3% (Bond et al. 2024). Early marine residence is considered a critical life stage for salmon, with factors such as size (Beamish et al. 2004), timing of marine entry (Scheuerell et al. 2009; Satterthwaite et al. 2014), and early ocean

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growth (Tomaro et al. 2012; Miller et al. 2014) influencing survival and adult return abundance. Earlier migration and rapid growth upon ocean entry are traits often linked to higher survival.

Within the Columbia River Basin there are recovery and production goals for wild and hatchery populations. One key metric for evaluating salmonid survival is Smolt-to-Adult-Return rate (SAR), which estimates survival based on the proportion of PIT-tagged (Passive Integrated Transponder) smolts leaving a specified location that return to that location as adults. For both wild and hatchery ICRSS Chinook salmon, target recovery SARs are often not met. The Northwest Power and Conservation Council set a recovery range of 2-6% overall SARs for wild spring/summer Snake River Chinook salmon populations (Northwest Power and Conservation Council 2020). Hatchery-specific goals for SARs are lower than for wild populations; for example, the target range for overall SARs at hatcheries within the Lower Snake River Compensation Plan is 0.38–0.97% (Northwest Power and Conservation Council 2023).

The Juvenile Salmon Ocean Ecosystem Survey (JSOES), conducted by the National Oceanic and Atmospheric Administration (NOAA) since 1998, provides a valuable long-term dataset that allows for detailed investigations into salmonid survival and recruitment dynamics (Losee et al. 2014; Daly et al. 2017; Litz et al. 2019). This annual survey is designed to develop a mechanistic understanding of how trophic dynamics and conditions in the Columbia River plume and coastal ocean affect survival of juvenile salmonids (Morgan et al. 2025). The increased usage of parentage-based tagging (PBT), a genetic method linking individual fish to their hatchery of origin (Steele et al. 2019; Hargrove et al. 2021), has allowed us to examine JSOES catch data in unprecedented detail. Given that 90% of the recent JSOES catch of ICRSS Chinook salmon were from hatcheries and approximately 70% of these fish are identifiable to hatchery of origin using

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PBT (Table 1), this dataset offers a unique opportunity to explore post-release performance across hatcheries.

Hatchery practices are known to affect post-release performance, influencing both physiological development (Beckman et al. 2017) and migration timing, with larger yearlings at release migrating seaward more quickly than smaller ones (Connor et al. 2004). The goal of our study was to examine migratory survival traits of yearling ICRSS Chinook salmon from different hatcheries and wild populations within and across years. Therefore, using JSOES collections of ICRSS Chinook salmon from 2015–2019 and 2021, we determined: (1) how hatchery composition of the JSOES catch compares to hatchery releases, (2) if size and condition at capture, marine entry timing, size at marine entry, and early ocean growth varied across hatcheries and between hatchery and wild yearlings, and (3) whether hatchery-specific variation in size and condition at capture can explain intra-annual variation in SARs. By investigating how hatchery of origin shapes early marine residence traits and marine survival, we aim to inform hatchery management and support the ecological resilience of ICRSS Chinook salmon populations.

Methods

Sample Collection

Fish were collected during NOAA's annual Juvenile Salmon Ocean Ecosystem Survey (JSOES). All research was carried out in accordance with applicable guidelines and regulations under Oregon State University's Institutional Animal Care and Use Committee (IACUC) protocols (Animal Welfare Assurance Number D16-00145, IACUC reference numbers 4791 and 5144). Sampling stations ranged from the mouth of the Columbia River (46.14°N) to northern

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Washington (47.93°N; Fig. S1). Surveys are currently conducted twice a year, in May and June, but our study focuses specifically on the May survey, when the vast majority of ICRSS Chinook salmon are collected. Salmonids were collected during daylight hours using a Nordic 264 pelagic rope trawl fished at the surface directly astern of the F/V *Frosti*, traveling at a speed over ground of approximately 6 $km \cdot h^{-1}$ (3.2 knots) for 30 minutes. The salmon collected by this survey reflect the population after any estuarine and early ocean mortality occurred. All salmon were sorted by species using standard taxonomic keys, enumerated, measured (fork length, FL, to the nearest mm), and a fin clip was taken and placed in ethanol until genetic analysis could be run. Fish were then frozen and stored until further processing in the laboratory. The juvenile salmon were later thawed, re-measured for length and mass, and sagittae (referred hereafter as "otoliths") were removed and placed in ethanol. No survey took place in 2020.

Genetic Analysis

Genetic stock identification (GSI) was used to gain information on the lineage of individuals and parentage-based tagging (PBT) was used to determine specific hatchery of origin in our catch (Teel et al. 2014). GSI uses genotyped samples from potential source populations and groups them into genetic stocks, providing information on relative abundance of different stocks within a system (Van Doornik et al. 2024). PBT is a genetic tool used to determine the origin of individuals using DNA (Steele et al. 2019). At several hatcheries, broodstock are sampled annually to create a baseline, and offspring can then be genetically assigned to their parents. Single Nucleotide Polymorphism Program for Intergenerational Tagging was used to infer parentage based on single nucleotide polymorphisms. Unlike other tagging methods, such as coded wire tagging, PBT does not involve any physical tagging, which makes it easier to track

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fish without the risk of tag loss. Occasionally, however, fish from one hatchery are transferred to a different hatchery for rearing and release. According to available hatchery release data, 27% of the fish from Little White Salmon National Fish Hatchery (NFH), located in the Mid-Columbia region, were transferred to Clearwater Hatchery, located in the Snake River, for rearing and release during the years of this study. This was the only known case in our study where the PBT assignment indicated one hatchery of origin, but the fish were raised and released from another hatchery.

Survey design and post-hoc sub-sampling varied across years. From 2015 to 2021 the main objective of the JSOES May survey was to maximize catch of ICRSS Chinook salmon (Fig. S1). Prior to 2021, we randomly subsampled fish from the catch across all stations because our intent was to examine migration patterns of the overall population of ICRSS Chinook salmon. However, as PBT became more prevalent, we observed that intra-annual variation in size at capture was related to PBT origin. Therefore, in 2021 we selected PBT fish only from hatcheries that had five or more fish in the catch to provide a more robust intra-annual comparison across those hatcheries.

Lastly, for our study, we use the term “wild” for untagged and unmarked individuals within our catch, although this designation does not confirm that these fish are truly wild. Our wild group is an upper limit, likely to include hatchery fish with no adipose fin clip. Virtually all hatchery fish are marked, however approximately 0.1% are released unmarked due to errors in the fin clipping process (Brandt et al. 2022).

Composition of JSOES Catch

To qualitatively assess how the composition of our catch compared with the composition of hatchery releases, we collated the numbers of ICRSS hatchery fish caught during the JSOES

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cruise, based on PBT data, and the numbers released for those hatcheries (Fish Passage Center 2025; Regional Mark Processing Center 2025). We included hatcheries from which more than five fish were caught over the course of two or more years (Table 2). First, we grouped PBT hatchery yearlings by their respective basin: seven hatcheries in the Mid-Columbia River, four in the Upper Columbia River, and eight in the Snake River (Fig. S2). Walla Walla Hatchery was assigned to the Mid-Columbia in alignment with NOAA's most recent ESU status report (Ford 2022). We then calculated the proportion of the catch (P_{ihc}) and the proportion of the release (P_{ihr}) for the hatcheries represented in the JSOES catch within each of the three basins for each year:

$$P_{ihc} = \sum_{j=1}^{n_i} C_{ij} / \sum_{k=1}^{3} \sum_{j=1}^{n_k} C_{kj}$$

$$P_{ihr} = \sum_{j=1}^{n_i} R_{ij} / \sum_{k=1}^{3} \sum_{j=1}^{n_k} R_{kj}$$

where:

C_{ij} = catch from hatchery j in basin i

R_{ij} = releases from hatchery j in basin i

n_i = number of hatcheries in basin i

n_k = number of hatcheries in basin k (for k =1, 2, 3, corresponding to the three basins)

We determined the proportions of catch (P_{hcwb}) and release (P_{hrwb}) from each hatchery within each basin for each year, which allowed us to examine representation within basins.

$$P_{hcwb} = C_{ij} / \sum_{j=1}^{n_i} C_{ij}$$

$$P_{hrwb} = R_{ij} / \sum_{j=1}^{n_i} R_{ij}$$

where:

C_{ij} = catch from hatchery j in basin i

R_{ij} = releases from hatchery j in basin i

n_i = number of hatcheries in basin i

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We then subtracted the proportion of the total released from the proportion caught to indicate whether hatchery production was over- or under-represented in our catch compared to releases across basins and then within basins. Given the numerous causes for variation in timing of migration and in-river survival that were not accounted for in our analysis, our intent was to present a qualitative comparison between our catch and hatchery releases.

Marine Entry Timing, Size, and Ocean Growth

We determined timing of marine entry, size at marine entry, and ocean growth for each fish using otolith structural and chemical analyses. Each otolith was mounted on a glass slide with Crystalbond™ 509 thermoplastic resin and initially polished with the sulcul groove (proximal) side facing upwards with wet-dry paper (1200-10000 grit) and alumina polish (0.3 µm) using an adjustable speed, multi-wheel polisher (Strait et al. 2025). Otoliths were then flipped and polished until daily increments became clear and then mounted randomly onto a shared glass slide in preparation for chemical analysis. Slides were cleaned ultrasonically in nano-pure water for 15 minutes, dried in a Class 100 hood, and stored in clean plastic bags until chemical analysis.

We used otolith strontium:calcium (Sr:Ca) and barium:calcium (Ba:Ca) ratios to determine the timing of marine entry. The increase in Sr:Ca and decline in Ba:Ca signifies freshwater emigration (Miller et al. 2010). Sr:Ca and Ba:Ca ratios were obtained through laser ablation inductively coupled plasma mass spectrometry (Thermo Scientific iCap RQ quadruple ICP-MS coupled to Photon Machines Analyte G2 laser) at Oregon State University's Keck Collaboratory for Plasma Mass Spectrometry, Corvallis, Oregon. The laser was set with a pulse rate of 10 Hz with a 30-micron spot size and traveled at a speed of 15 microns/sec from the otolith's ventral edge to the dorsal edge, crossing the core primordia. National Institute of

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Standards and Technology (NIST) 612 elemental glass standards were sampled throughout the session to calculate machine precision. Otolith analyte count per second data was normalized by ^{43}Ca to account for variation in instrument sensitivity and the amount of ablated material. These were then converted to elemental ratios based on NIST 612 values. Molar ratios for Sr:Ca (mmol/mol) and Ba:Ca (μmol/mol) were used to determine otolith width at freshwater emigration, hereafter referred to as "marine entry" (Tomaro et al. 2012; Miller et al. 2014). Precision (% Relative Standard Deviation, or RSD) was determined based on repeated measurements of NIST 612 glass slides (^{43}Ca = 7.75% (0.9 SD), ^{86}Sr = 8.65% (0.6 SD), ^{138}Ba= 10.0% (0.9 SD)). Accuracy was determined based on repeated analysis of USGS calcium carbonate standards MACS-1 (mean ± SD for Sr:Ca = 1.01% ± 0.08, Ba:Ca = 0.76% ± 0.3).

After elemental data were collected, each otolith was imaged using a Levenhuk M1000 camera connected to a Lecia DM1000 compound microscope at both 40× and 400× magnification. Imaged otoliths were analyzed using Image ProPlus™ software (Media Cybernetics®). Daily growth increments deposited after marine entry, which represent the number of days the fish had spent in the marine environment, were then counted from point of marine entry out to the otolith edge. Each otolith was interpreted at least twice by two different readers. If counts varied by >10%, the otoliths were re-read, and conflicts were resolved.

We estimated marine entry date by subtracting the number of days in the ocean from the date of capture. It can take up to 2 to 3 days for the elevated strontium associated with marine waters to be reflected in the otolith transect (Miller 2011). Therefore, samples with no elevated Sr:Ca were randomly assigned a time of marine entry of 2 or 3 days prior to catch. We also collated basin specific PIT-tag detections of spring/summer smolts crossing Bonneville Dam (Columbia River DART; www.cbr.washington.edu/dart) to compare with our estimated dates of

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marine entry. Otolith size is often used as a reliable proxy for estimating fish size (Campana and Neilson 1985); therefore, we used otolith size as a proxy for length at marine entry. Daily growth rates were estimated by subtracting otolith size at capture from otolith size at marine entry and divided by estimated ocean residence time.

Comparison of Fish Metrics by Origin

We examined sample size vs. standard deviation plots to determine adequate sample sizes to quantify length, condition, and growth for each year x hatchery combination, and standard deviations plateaued between 4 and 6 samples. Therefore, we included hatcheries that were represented in at least two years with $n \geq 5$ in each year (Table 3) and years with data for at least two hatcheries in subsequent analyses. We used linear (LM) and linear mixed models (LMM) with and without collection year as a random effect to determine if length and condition at capture, timing of marine entry, otolith size at marine entry, and otolith marine growth varied across origins. For length and condition at capture, we included fish from 13 hatcheries, as well as all wild fish. For timing of marine entry, otolith size at marine entry, and otolith marine growth, we included 11 hatcheries and wild fish. For some hatcheries and years, an overall mean size at release (or tagging) was available (Table S1). Therefore, we also compared reported mean length at release with mean length at capture and with estimated otolith size at marine entry in year x hatchery combinations with >5 fish.

For all models, the optimal model structure was identified for both random and fixed effects using Akaike Information Criteria (AIC) and maximum likelihood (ML). When random effects were included, the final models were estimated using restricted maximum likelihood (REML). Model fit was evaluated through examination of the normality, variance, and linearity of

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the residuals and we evaluated multicollinearity with Variance Inflation Factor (VIF), when appropriate. Marginal model means were generated with the R packages 'effects' (Fox and Weisberg 2019) and 'ggeffects' (Lüdecke 2018). Models were developed using the R package 'lme4' (version 1.1.31) (Bates et al. 2015). In addition to the model results, we reviewed Tukey post-hoc pairwise comparisons via the 'emmeans' R package. All statistical analyses were conducted in RStudio 2024.04.1+748 with R version 4.4.2.

Variation in Marine Survival

There are various estimates of "survival" within the Columbia River basin. We focused on marine survival and used smolt-to-adult return (SAR) estimates, which are based on the proportion of smolts leaving a specified location that return to that location as adults using PIT tag detections (ISAB 2025). We compared SAR estimates based on survival from Bonneville Dam back to Bonneville Dam (BON-BON), which is the lowermost dam on the Columbia River (Fig. 1). For our survival analysis, we included hatcheries that were represented in the JSOES catch and for which hatchery-specific SARs were available: Carson NFH [Mid-Columbia], Leavenworth NFH and Winthrop NFH [Upper Columbia], and Clearwater Hatchery, Dworshak NFH, McCall Hatchery, Pahsimeroi Hatchery, Rapid River Hatchery, and Sawtooth Hatchery [Snake].

For Mid-Columbia hatcheries, SARs were available for BON-BON. For Snake River hatcheries, hatchery-specific SARs were available for Lower Granite Dam (LGD), the uppermost Snake River dam with fish passage to BON. For Upper Columbia hatcheries, hatchery-specific SARs were available for McNary Dam (MCN) to BON. Therefore, we used annual estimates of in-river juvenile survival to BON to adjust the Snake River and Upper Columbia River SARs to BON-BON (McCann et al. 2024; Widener et al. 2024). We also adjusted Snake River in-river

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247 survival for the proportion of juveniles that were transported via barge from LGD to below BON.
248 These adjustments allowed us to compare estimates of $SAR_{BON\text{-}BON}$ across all hatcheries
249 (Supplemental Information). $SAR_{BON\text{-}BON}$ were transformed using the logit function for statistical
250 analysis to meet model assumptions.

251 Within the Columbia River Basin, previous studies (Tomaro et al. 2012; Miller et al.
252 2014) reported positive correlations ($r > 0.85$) between annual variation in mean length at capture
253 for JSOES yearlings and survival proxies from 1999-2008. From 2000-2021, correlations between
254 annual mean length at ocean capture for all yearlings and $SAR_{BON\text{-}BON}$ remain positive ($r = 0.58 –$
255 0.62), although weaker than earlier years. With the increased usage of PBT, we are now able to
256 investigate the variation in survival within years *after* taking year into account. Therefore, we
257 compared $SAR_{BON\text{-}BON}$ using linear mixed models with hatchery of origin and length or condition
258 at ocean capture as fixed effects and outmigration year as a random factor. We included $SAR_{BON\text{-}BON}$ estimates for migration years 2015, 2016, 2018, 2019 and 2021; 2017 was removed because
260 we only collected fish from one hatchery with more than five fish. $SAR_{BON\text{-}BON}$ were estimated
261 with and without precocious males, or jacks. We present models for hatchery-specific $SAR_{BON\text{-}BON}$ estimates without jacks, but the results were very similar for $SAR_{BON\text{-}BON}$ estimates with
263 jacks, likely related to the strongly positive correlation between the adjusted $SAR_{BON\text{-}BON}$
264 estimates with and without jacks ($r = 0.98$).

265 Various factors associated with each hatchery could influence their relative survival.
266 Therefore, we evaluated some of the hatchery-specific variables associated with migration,
267 including distance to the ocean and number of dams passed during migration. We compared our
268 best LMM predicting $SAR_{BON\text{-}BON}$, without jacks based on hatchery of origin, with models that
269 substituted hatchery of origin with hatchery distance to the ocean (km) or the number of mainstem

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dams between the hatchery and ocean. Given high collinearity, models with all three variables were not supported, so we used this comparative approach to compare the relative variation in $SAR_{BON\text{-}BON}$ explained by these variables. ICRSS Chinook salmon have very low ocean intercept rates; however, Columbia River fisheries are responsible for an incidental take of salmon returning to the Columbia River mouth that averages around 11 percent across the years of our study (NMFS 2022).

Results

Analysis of Catch

The direct comparison between the JSOES catch and hatchery release data revealed temporal variation in representation (Fig. 2). From 2015 through 2021, the survey caught yearlings assigned to 23 hatcheries based on PBT identification and some hatcheries consistently contributed to the overall catch, while others appeared irregularly over the years (Table 2). In 2017 and 2019, catch was low across all areas of the Columbia River Basin, which coincided with very low SARs. Snake River hatcheries were over-represented in the catch relative to hatchery releases by 6 to 24% in 2015, 2016, and 2019. Fish from Mid-Columbia River hatcheries were over-represented in 2017 and 2021 by 13 to 27% and under-represented in the other years. Fish from Upper Columbia River hatcheries were consistently under-represented but only by <7%. Hatchery release data can be found in Table S2.

Our comparisons of hatchery representation in the catch within basins provided additional detail (Fig. S2). In the Snake River, hatchery representation in the catch was the most variable. Clearwater Hatchery was often under-represented by 10 to 20% in 2015-2018, whereas Dworshak NFH was over-represented by ~15% in 2017 and 2021. For Mid-Columbia, over-representation at

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the basin level in 2017, 2019, and 2021 was primarily due to Carson and Little White Salmon NFH. In Upper Columbia, where JSOES catch and hatchery release aligned relatively well at the basin scale, Leavenworth NFH was over-represented in catches compared to other hatcheries, except for 2017 and 2021.

Marine Entry Timing

Approximately 26% of the variation in marine entry timing was described by the LMM with origin as a fixed effect and year as a random factor ($R^2_{Marginal} = 0.124$, $R^2_{Conditional} = 0.256$; Table S3, S3.1). In 2021, the only year with marine entry timing data available for Upper and Mid-Columbia hatcheries, most Mid-Columbia hatcheries entered first, followed by Upper Columbia hatcheries, and then Snake River hatcheries, except for one fish from Dworshak NFH, which entered the ocean before all other fish. In 2016 and 2021, entry timing shows well-defined peaks for most hatcheries, whereas in 2018 the entry date was more protracted (Fig. 3). For 2016 and 2021, wild fish entered throughout the emigration period, which is expected as they originated from throughout the interior basin. The comparison between timing of marine entry of JSOES fish with the timing of passage over Bonneville Dam, based on PIT tags, indicated that Mid-Columbia hatcheries completed their passage before the survey in 2021, but wild, Upper Columbia, and Snake River fish were in the middle of their peak passage timing during the survey. For 2016 and 2018, survey timing occurred at or near the end of peak smolt passage over Bonneville Dam.

Size at Marine Entry and Early Ocean Growth

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315 Origin accounted for approximately 45% of the observed variation in marine entry size
316 ($R^2_M = 0.454$, $R^2_C = 0.460$; Table S4, S4.1). Carson NFH fish were significantly smaller at entry
317 than fish from Dworshak NFH and Lookingglass Hatchery, but significantly larger than wild. In
318 2016, most wild fish entered the ocean at smaller sizes than hatchery fish, whereas in 2021, their
319 entry sizes had become more comparable to hatchery fish (Fig. 4). In 2021, fish from Chief
320 Joseph Hatchery entered the ocean at larger sizes compared to the others. There was no significant
321 difference in ocean growth by origin ($R^2_M = 0.061$, $R^2_C = 0.209$; Table S5, S5.1). In 2018, fish
322 exhibited a narrower range of early marine growth compared to 2016 and 2021, when growth
323 patterns were more variable (Fig. 5).
324
325 *Size and Condition at Capture*
326 Length at capture (FL, mm) was best explained by a linear model with origin as a fixed
327 effect and no random effect of year ($R^2 = 0.193$; Table S6, S6.1), which indicates there is little
328 residual interannual variability after accounting for origin effects. Yearlings from Carson NFH,
329 Little White Salmon NFH, Chief Joseph, Clearwater, Pahsimeroi, and Rapid River hatcheries
330 were consistently the largest and similar in length ($p < 0.05$; Fig. 6A), whereas wild individuals
331 were significantly smaller than all other represented hatcheries ($p < 0.05$), apart from Winthrop
332 NFH and Lookingglass Hatchery.
333 Inter-annual variation in condition at capture was more pronounced, necessitating the
334 inclusion of year as a random effect. The fixed factor of origin explained 13.1% of the variation in
335 condition at capture, and the overall model accounted for 27.4% ($R^2_M = 0.131$, $R^2_C = 0.274$, $p <$
336 0.0001, $n = 1055$; Table S7, S7.1). Carson NFH fish were in significantly higher condition than
337 all other hatchery and wild fish ($p < 0.05$; Fig. 6B), except when compared to other Mid-

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Columbia hatcheries, including Little White Salmon NFH and Parkdale Hatchery. In comparison, fish from the Snake River and Upper Columbia River hatcheries were in lower condition and similar to each other. Wild fish were in significantly higher condition at capture than Sawtooth, Rapid River, McCall, and Clearwater hatcheries ($p < 0.05$).

We also determined that hatchery-specific mean length at capture was positively related to mean length at release, indicating that fish that were larger at release tended to remain larger in the ocean (LMM with year as a random effect: $R^2_M = 0.114$, $R^2_C = 0.189$, $p < 0.0001$, $n = 36$; Fig. S3). Chief Joseph Hatchery fish in 2021 were notably larger (mean = 179.2 mm FL, $n = 7$) than other years or hatcheries. Excluding fish from Chief Joseph Hatchery in 2021, there was a stronger positive relationship between size at release and size at capture across all years ($R^2_M = 0.289$, $R^2_C = 0.441$, n = 38, $p < 0.0001$). Although we had fewer estimates of size at marine entry for comparison, we observed a positive correlation between size at marine entry and size at release, but only when Chief Joseph Hatchery fish in 2021 were removed ($r = 0.321$, $p = 0.204$, $n = 17$ versus $r = 0.722$, $p = 0.0007$, $n = 16$). However, reporting practices of size at release varied and often include no measurement of variance, limiting our ability to make certain comparisons.

Hatchery of Origin Influencing SARs

Annual mean hatchery-specific $SAR_{BON\text{-}BON}$ were positively correlated with annual mean hatchery-specific body condition ($r = 0.356$, $p = 0.031$, $n = 37$), but not with length ($r = -0.090$, $p = 0.59$, $n = 37$) or mass ($r = 0.021$, $p = 0.902$, $n = 37$). We examined how intra-annual variation in length and condition was related to $SAR_{BON\text{-}BON}$ across rearing hatcheries, after accounting for interannual variation, using LMMs. Our most parsimonious model included hatchery of origin, condition-at-capture, and their interaction as fixed effects and migration year as a random factor

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(Table S8). Although the interaction in the top model resulted in high collinearity, this was due to structural, not design, collinearity and was therefore retained. This top model accounted for 71.5% of the variation in $SAR_{BON-BON}$, the fixed effects accounted for 46.7%, and the random effect of year accounted for 24.8% of the variation (Table 4, S9).

Over the years, the ranking of $SAR_{BON-BON}$ across hatcheries remained consistent, as hatcheries located in the Mid- and Upper Columbia basin (e.g., Carson, Winthrop, and Leavenworth NFH) consistently displayed higher $SAR_{BON-BON}$ compared to hatcheries within the Snake basin (Fig. 7). Carson and Winthrop NFH consistently had the highest $SAR_{BON-BON}$ (marginal mean = 0.368 and 0.421 and overall mean = 0.431 ± 0.187 and 0.594 ± 0.187, respectively) whereas Pahsimeroi Hatchery had the lowest (marginal mean = 0.062 and overall mean = 0.071 ± 0.043). The effect of condition was not consistent across hatcheries. $SAR_{BON-BON}$ increased with body condition only at Leavenworth NFH and decreased with body condition for Pahsimeroi, Winthrop, and McCall hatcheries (Table 4, Fig. S4). When Model M3 was compared with alternative models that included condition and hatchery distance from the ocean or the number of mainstem dams, there was much greater support for Model M3. The fixed effects for all other models accounted for <17% of the variation in $SAR_{BON-BON}$ whereas Model M3 accounted for 46.7% (Table S10). The dam model (D2) indicated that survival declined with the number of dams passed during migration and there was an interaction ($p = 0.04$) with higher body condition having a weak positive effect on survival for fish passaging fewer than ~5 dams. Similarly, there was a positive effect of body condition on survival within ~1000 km of the ocean, but that relationship reversed at greater distances.

Discussion

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Hatchery effects on post-release performance have been documented (Harstad et al. 2018; Nelson et al. 2019); however, to our knowledge, this is the first study to examine intra-annual variation in juvenile migration patterns and early ocean residence across hatcheries based on PBT identification. We found persistent phenotypic differences in early marine residence traits and variation in marine survival among hatcheries, indicating that hatchery of origin can influence performance in the ocean weeks to months after release. Notably, hatchery of origin explained more of the variation in marine survival than fish length and condition at capture, dams crossed, and geographic location of the rearing facility. These findings suggest that hatchery-specific practices may be shaping marine survival in ways not captured by fish condition. Identifying the drivers regulating these observed patterns could contribute to the development of more effective management strategies to support the sustainability of salmon populations.

Juvenile salmon collected by JSOES reflect ICRSS Chinook salmon populations *after* any estuarine and early ocean mortality occurred, which is often referred to as a critical period with high and variable mortality (Tomaro et al. 2012; Norrie et al. 2022). The relative composition of the JSOES catch compared to hatchery release were reasonably similar (within 27% of each other throughout the years) indicating that the May survey provides an informative snapshot of released hatchery fish from the Columbia River Basin. The catch composition was, however, influenced by the survey timing relative to ICRSS Chinook salmon migration patterns. Estimates of migration timing from Bonneville Dam to the ocean are often 3 to 5 days (Weitkamp et al. 2022). Comparing survey timing to smolt PIT tag detections at Bonneville Dam revealed that the 2016 and 2018 surveys occurred near the end of peak smolt migration for Snake River and wild populations. However, the 2021 survey occurred towards the middle of smolt migration for all groups except the Mid-Columbia, suggesting that later migrants from hatcheries in the Upper

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Columbia and Snake River basin, as well as wild fish, may not be represented in the catch. Comparison of catch composition with hatchery releases supports this conclusion as Mid-Columbia hatcheries were over-represented while Snake hatcheries were under-represented in 2021. Although migration timing can fluctuate based on environmental factors (Achord et al. 2007), the mean date for 50% passage over Bonneville Dam of Upper Columbia and Snake River hatchery fish in 2021 (21 May) was very similar to the 15-year mean (20 May) (Columbia Basin Research 2025). A more comprehensive analysis of JSOES catch in relation to PIT tag detections of out-migrating smolts at Bonneville Dam could provide a framework for tailoring future survey timing to maximize data richness for specific stocks.

Marine entry timing has been linked to salmonid survival, with earlier entry generally associated with higher survival rates (Chasco et al. 2021; James et al. 2023; Bond et al. 2024). In our study, migration timing was best explained by year and origin equally, each accounting for ~12%. Migration patterns are often attributed to hatchery release strategies, such as the number of fish released and the method of release (Duffy and Beauchamp 2011; Weitkamp et al. 2015). However, release strategies during the time of this study remained relatively consistent across facilities (R. Engle, pers. comm.). The variation in timing we observe is then likely due to hatchery-specific differences associated with facility location such as river flow (Notch et al. 2020), water temperature (Zydlewski et al. 2005; Burke et al. 2013), or social cues (Kururvilla et al. 2024).

Large body size at marine entry and during early ocean residency are also known to provide a survival advantage (Duffy and Beauchamp 2011; Woodson et al. 2013) likely because larger individuals are better able to compete for resources (Reinhardt et al. 2001; Gregory et al. 2019). In our study, origin accounted for approximately 45% of the variation in size at marine

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entry, supporting the idea that size at release strongly influences size at marine entry (Weitkamp et al. 2015). Size at marine entry was consistent with size at capture, with individuals that entered the ocean at larger sizes generally remained larger during early marine residence. Notably, early ocean growth rates were similar across hatcheries within years, suggesting that size at and timing of ocean entry were the primary drivers of the variation in hatchery-specific size at capture, rather than differential growth. The interannual variation in early ocean growth may then reflect broader environmental fluctuations, including water temperature (Marine and Cech 2004), upwelling strength (Logerwell et al. 2003), or available prey (Pool et al. 2012).

Long-term research at the stock level report positive relationships between body size, condition, and survival across years (Litvak and Leggett 1992; Gregory et al. 2019). Our study further explored how survival was related to intra-annual variation in size and condition across hatcheries. We determined that variation in length at capture was consistent enough across years where year effects were not retained in our model, further supporting the strong influence of hatchery of origin on early ocean size. In contrast, condition at capture required year to be included, suggesting that factors unique to each year may play a larger role in shaping condition than size. Together, these results highlight how size and condition may be informative predictors of survival at broad temporal scales yet provide limited explanatory power for intra-annual variation in survival.

We observed variable relationships between $SAR_{BON\text{-}BON}$ and migration size-related traits across hatcheries. Carson NFH entered the marine environment earliest and exhibited high survival, whereas Pahsimeroi Hatchery and Dworshak NFH also entered early but had the lowest survival rates. Average size at ocean entry likewise failed to explain variation in $SAR_{BON\text{-}BON}$, as Carson and Dworshak NFH entered the ocean at similarly small sizes but had opposing survival

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453 outcomes. Body size at capture also showed no consistent relationship with survival. Winthrop
454 NFH, which exhibited the highest $SAR_{BON-BON}$, produced the smallest fish at capture, whereas
455 Carson NFH was among the largest. Conversely, Pahsimeroi Hatchery, which had the lowest
456 survival, produced the largest fish, while Dworshak NFH produced some of the smallest. Though
457 hatchery-specific $SAR_{BON-BON}$ was positively correlated with condition at capture, the patterns
458 were inconsistent. Winthrop NFH exhibited higher survival at lower condition while Leavenworth
459 NFH and Pahsimeroi Hatchery showed higher survival at higher condition. Collectively, these
460 results indicate no clear within-year survival advantage related to timing, size, or condition in our
461 data set, which focuses on yearlings that survived their early marine residence. Our findings
462 instead point to likely carry-over effects, defined as events or processes that influence individual
463 condition or performance later in time (Harrison et al. 2011; Moore and Martin 2019), associated
464 with hatchery of origin.

465 Carry-over effects can affect vital rates such as growth or survival (Araki et al. 2009). The
466 ranking of $SAR_{BON-BON}$ across hatcheries in our study remained consistent throughout the years.
467 Snake River hatchery fish, which consistently had the lowest $SAR_{BON-BON}$, are speculated to
468 experience delayed mortality, where cumulative stress or injury from dam passage increases
469 mortality later in the estuary or early ocean residence phase (Schaller and Petrosky 2007).
470 However, this hypothesis remains poorly understood and some studies find no evidence to
471 support delayed mortality (Rechisky et al. 2013). Consistent with this uncertainty, hatchery of
472 origin explained substantially more of the variation in $SAR_{BON-BON}$ than the number of dams or
473 distance between the hatchery and ocean. Our results point toward hatchery-specific factors, such
474 as genetic or broodstock differences, late-stage mortality, or release method, timing, and location,
475 as likely drivers of the observed carry-over effects. Carson NFH stock, with some of the highest

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survival rates, originated from collections of spring Chinook salmon at Bonneville Dam in the 1950s (Ford et al. 2004) and has since been distributed among hatcheries throughout the Columbia River Basin. Spring-run Chinook salmon released from Leavenworth, Little White Salmon, and Winthrop NFH are primarily Carson-origin stock (Myers et al. 1998; Maier 2017). This could partially explain why Mid-Columbia hatcheries tend to outperform other basins; however, many Snake River hatcheries also received significant contributions from Carson-stock hatcheries (Matthews and Waples 1991). This pattern aligns with research showing that the same stock can exhibit major differences in performance when reared at different facilities (Bumgarner et al. 2023).

Release location may further modify survival outcomes, as stocks reared under the same conditions but released in different locations have demonstrated divergent survival rates (Feldhaus et al. 2022). While hatchery rearing practices are influencing smolt performance post-release, the timing and location of release adds an additional selective pressure. Size at release may further interact with release location to influence survival. Larger fish may face increased predation risk due to their increased visibility to predators such as seabirds, larger fish, or marine mammals (Collis et al. 2001, Hostetter et al. 2023). In areas of high predation, it may be beneficial to be smaller. Migration timing can also influence predator-prey overlap (Källo et al. 2020), such that even within the same basin, variation in release timing and location could expose smolts to different predation pressures, regardless of size. Upon return to the Columbia River as adults, predation by marine mammals is estimated to account for 20-44% mortality (Wargo Rub et al. 2019). Collectively, these highlight the complexity of survival relationships and the need for a greater understanding of factors influencing post-release survival across specific hatcheries.

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In the Columbia River, hatchery ICRSS Chinook salmon overwhelmingly dominate the system, with presumptively wild fish making up less than 10% of the JSOES catch. While hatchery programs are an integral component of the Columbia River, there are concerns about potential negative impacts on wild Chinook salmon populations, such as reduced genetic diversity and lower fitness of hybrid offspring (Araki et al. 2007; Grant 2012). Consistent with previous studies, we found wild fish were smaller at capture when compared to hatchery fish (Nickelson 2003; Weber and Fausch 2003; Beamish et al. 2012; Bottom et al. 2021) and greatly varied in their marine entry timing. Despite their smaller size, wild fish were in better condition than most of the hatchery fish, possibly reflecting adaptations that enhance survival in variable environments (Jonsson et al. 2003; Claiborne et al. 2014). Given the substantial presence of hatchery fish in the marine environment, it is essential to evaluate how yearlings from different hatcheries are responding to the shared environment after release, while also considering the ecological role and state of the remaining wild population.

Our analysis provides a comprehensive review of the factors regulating relative marine survival in the Columba River Basin. While improvements to fish passage infrastructure in the Columbia River Basin since the early 1990s have increased juvenile and adult survival (Williams et al. 2005), many current hatchery practices are based on earlier studies prior to improvement in passage conditions and before the more recent increases in temperature in both freshwater and marine salmon habitats. Given the consistent differences in marine survival of ICRSS Chinook salmon, it would be worthwhile to reassess these practices in order to optimize the ecological health of their stocks. A basin-wide analysis of pre- and post-release performance could also further elucidate the mechanisms behind the persistent survival variations across hatcheries. Unravelling how these complex hatchery dynamics interact to influence marine survival will be a

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521 critical component of the management and long-term sustainability of Interior Columbia River
522 spring/summer Chinook salmon.
523

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537 Competing Interests

538 The authors declare there are no competing interests.
539

540 Data Availability

541 Data and code used in this study are publicly available through GitHub at
542 https://doi.org/10.5281/zenodo.18486669.

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Table 1. Percentage of all Interior Columbia River spring/summer Chinook Salmon captured in the May Juvenile Salmon Ocean Ecosystem Survey (JSOES) that were tagged. "Tagged" includes adipose fin clip, parentage-based tagging (PBT), coded wire tags, Passive Integrated Transponder tags, dye elastomer tags, or any combination thereof. Percent of fish PBT includes *only* fish with PBT data. No survey was conducted in 2020.

	Survey Year						
	2015	**2016**	**2017**	**2018**	**2019**	**2021**	**Total**
Total fish caught	245	379	36	336	75	367	1,438
Fish tagged	234	348	29	310	72	342	1,335
Percent of fish tagged	95.5%	91.8%	80.6%	92.3%	96.0%	93.2%	92.8%
Percent of fish PBT	76.7%	81.5%	66.7%	67.0%	80.0%	75.7%	77.9%

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Table 2. Number of yearling Interior spring/summer Chinook salmon caught in the May Juvenile Salmon Ocean Ecosystem Survey (JSOES) with PBT data and originating from Columbia River hatcheries. No survey took place in 2020. Hatcheries with an "*" are included in the Evolutionarily Significant Unit (ESU) listed as "Threatened" under the Endangered Species Act. "C" indicates hatcheries included in catch versus release comparison. "NFH" refers to National Fish Hatchery.

Basin	Hatchery	2015	2016	2017	2018	2019	2021	Total
Mid-Columbia River	Carson NFH[C]	10	12	6	24	5	46	**103**
	Cle Elum Hatchery[C]		3	1			4	**8**
	Klickitat Hatchery[C]	2			12		1	**15**
	Little White Salmon NFH[C]	3	6	4		4	58	**75**
	Parkdale Hatchery[C]	2	2	1	5	1	8	**19**
	Round Butte Hatchery	2			2			**4**
	Umatilla Hatchery[C]	2	1		3			**6**
	Walla Walla Hatchery						11	**11**
	Warm Springs NFH[C]		10	1	1			**12**
Upper-Columbia River	Chief Joseph Hatchery[C]		2		8		7	**17**
	Leavenworth NFH[C]	10	17		16	3	6	**52**
	Methow Hatchery*[C]	2	7					**9**
	Winthrop NFH*[C]	15	3		2	2	7	**29**
Snake River	Clearwater Hatchery[C]	9	34		5	15	35	**98**
	Dworshak NFH[C]	8	21	3	24	6	37	**99**
	Lookingglass Hatchery*[C]	14	25		9			**48**
	Lyons Ferry Hatchery	3						**3**
	McCall Hatchery*[C]	31	42	1	19	6	9	**108**
	Nez Perce Hatchery[C]	2	2		4			**8**
	Pahsimeroi Hatchery*[C]	3	35	1	19	2	1	**61**
	Powell Hatchery		1	2			2	**5**
	Rapid River Hatchery[C]	49	68	3	47	14	39	**220**
	Sawtooth Hatchery*[C]	21	18	1	25	2	7	**74**
Unmarked		15	35	7	34	5	29	**125**
Total		**203**	**344**	**31**	**259**	**65**	**307**	**1,209**

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Table 3. Sample size for each analysis completed. Capture length and capture condition included all fish caught during the 2015-2019 and 2021 Juvenile Salmon Ocean Ecosystem Survey (JSOES) May surveys. SARs (smolt-to-adult returns) includes hatcheries that were represented in the JSOES catch for which hatchery-specific Bonneville to Bonneville SARs were available. Marine entry size and daily growth analyses included fish with a marine chemical signature present in the otolith. Marine entry date included fish with and without marine chemical signature in the otolith. "-" indicates no samples. “NFH” refers to National Fish Hatchery.

Basin	Hatchery	Capture Length & Condition	SARs	Marine Entry Size	Marine Growth	Marine Entry Date
Mid-Columbia River	Carson NFH	103	103	34	31	32
	Little White Salmon NFH	64	-	37	35	36
	Parkdale Hatchery	13	-	-	-	-
	Walla Walla Hatchery	-	-	8	7	7
Upper Columbia River	Chief Joseph Hatchery[S]	15	-	5	5	6
	Leavenworth NFH	48	51	-	-	-
	Winthrop NFH	22	29	-	-	-
Snake River	Clearwater Hatchery	98	98	24	10	27
	Dworshak NFH	96	99	31	26	35
	Lookingglass Hatchery	47	-	10	9	10
	McCall Hatchery	107	107	33	10	33
	Pahsimeroi Hatchery	54	60	36	23	36
	Rapid River Hatchery	217	220	68	40	78
	Sawtooth Hatchery	71	74	29	12	29
Wild	Unmarked	100	-	45	32	49

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Table 4. Results from linear mixed effect model (estimated using REML and nloptwrap optimizer) to examine the effect of hatchery of origin and condition on Bonneville to Bonneville Smolt-to-Adult return. Marginal $R^2 = 0.467$ and conditional $R^2 = 0.715$.

Predictor	**Estimates (SE)**	***p***
Intercept - Carson	-5.600 (0.167)	<0.001
Clearwater	-0.681 (0.067)	<0.001
Dworshak	-1.075 (0.066)	<0.001
Leavenworth	-0.629 (0.076)	<0.001
McCall	-0.684 (0.067)	<0.001
Pahsimeroi	-1.792 (0.076)	<0.001
Rapid River	-0.728 (0.060)	<0.001
Sawtooth	-0.591 (0.071)	<0.001
Winthrop	0.133 (0.102)	0.019
Condition	0.337 (0.345)	0.324
Clearwater x Condition	-0.197 (0.510)	0.695
Dworshak x Condition	-0.116 (0.490)	0.809
Leavenworth x Condition	1.762 (0.569)	0.002
McCall x Condition	-1.048 (0.471)	0.024
Pahsimeroi x Condition	-1.809 (0.763)	0.017
Rapid River x Condition	-0.062 (0.426)	0.881
Sawtooth x Condition	0.896 (0.504)	0.074
Winthrop x Condition	-1.638 (0.788)	0.036
Random Effects		
σ2	0.13	
Year	0.10	
ICC	0.44	
Nyear	5	
Observations	810	

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Figure 1: Map of the Columbia and Snake River basin showing location of hatcheries and major hydroelectric dams. Hatcheries are grouped and colored by basin (Purple = Mid-Columbia, Blue = Upper Columbia and Green = Snake) and dams are indicated by colored triangles. The Columbia mainstem is shown in light blue, the Snake in dark green, and the Salmon River in dark blue. Base map created using package 'maps' version 3.4.2.1 in RStudio.

Figure 2: The proportional difference between the Juvenile Salmon Ocean Ecosystem Survey (JSOES) catch and hatchery release data for Mid-Columbia (purple), Upper-Columbia (blue), and Snake River (green) spring/summer Chinook salmon yearlings caught in May. Hatchery release data were compiled from the Fish Passage Center [fpc.org] and Regional Mark Information System [rmpc.org]. Values above the dotted line (y=0) indicate basins overrepresented in the JSOES catch and below the dotted line are basins that are underrepresented in the JSOES catch, relative to hatchery release numbers. No JSOES survey occurred in 2020.

Figure 3: Marine entry timing (salinity > 10) of yearling Interior Columbia River spring/summer Chinook salmon caught during the May Juvenile Salmon Ocean Ecosystem Survey. Vertical dashed lines indicate the survey dates for each year. Density curves represent hatchery-specific distributions of marine entry day of year. Colors indicate the basin of each hatchery: purple (Mid-Columbia), blue (Upper Columbia), green (Snake River), and yellow (Wild). The colored rectangular boxes at the bottom of each panel represent the period during which 95% of Passive Integrated Transponder (PIT) tagged spring/summer smolts from each basin crossed Bonneville Dam.

Figure 4: Otolith size (μm) at marine entry for yearling Interior Columbia River spring/summer Chinook salmon caught during the May Juvenile Salmon Ocean Ecosystem Survey. Colors indicate the basin of each hatchery: purple (Mid-Columbia), blue (Upper Columbia), green (Snake River), and yellow (Wild).

Figure 5: Daily otolith growth (μm) during early marine residence for yearling Interior Columbia River spring/summer Chinook salmon caught during the May Juvenile Salmon Ocean Ecosystem Survey. Colors indicate the basin of each hatchery: purple (Mid-Columbia), blue (Upper Columbia), green (Snake River), and yellow (Wild).

Figure 6: Predicted length at capture (fork length, mm; **A)** and condition at capture (Fulton's K = mass/(length3); **B)** based on origin for hatchery and wild yearling Interior Columbia River spring/summer Chinook salmon caught in May. Error bars represent 95% confidence intervals. The letters above hatcheries represent post-hoc pairwise comparisons. Shared letters indicate hatcheries that are not significantly different ($p < 0.05$). Colors indicate the basin of each

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hatchery: purple (Mid-Columbia), blue (Upper Columbia), green (Snake River), and yellow (Wild).

Figure 7: Predicted back-transformed mean (±95% confidence intervals) Bonneville-Bonneville smolt-to-adult returns (SARs) for yearling Interior Columbia River spring/summer Chinook salmon from various hatcheries. The linear mixed model included condition at capture, hatchery of origin, and their interaction as fixed effects and year as a random effect. There was no survey in 2020 and the 2017 survey was removed because there was only one hatchery with more than five fish collected. Colors indicate the basin of each hatchery: purple (Mid-Columbia), blue (Upper Columbia), and green (Snake River).

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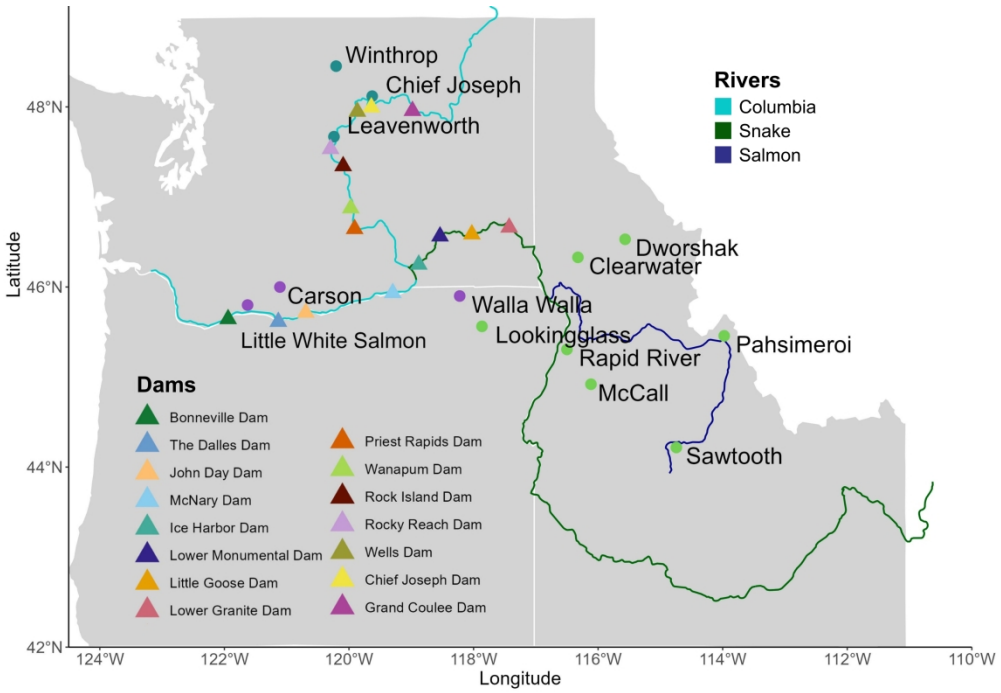


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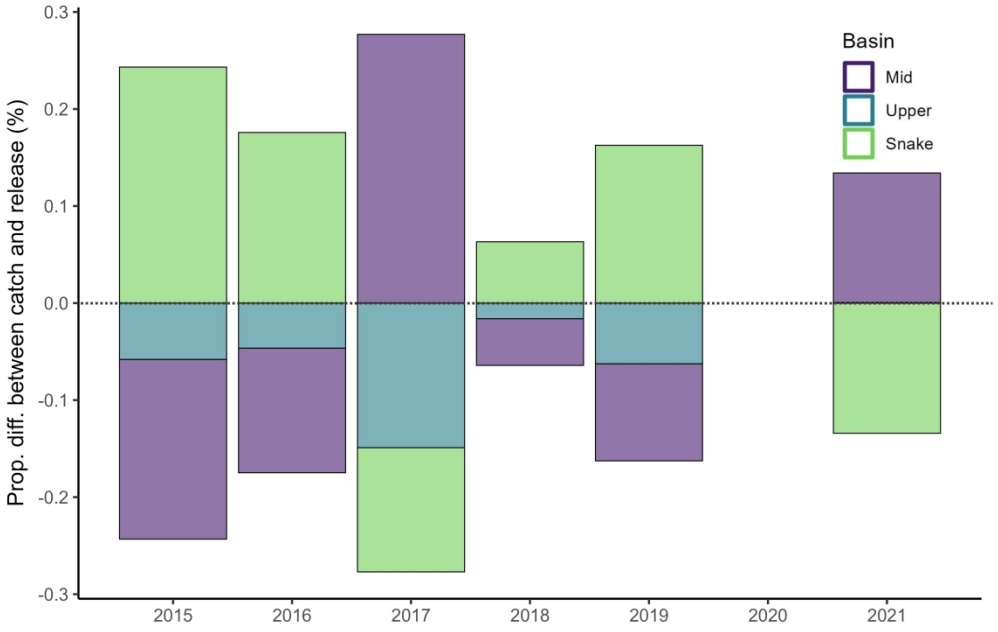


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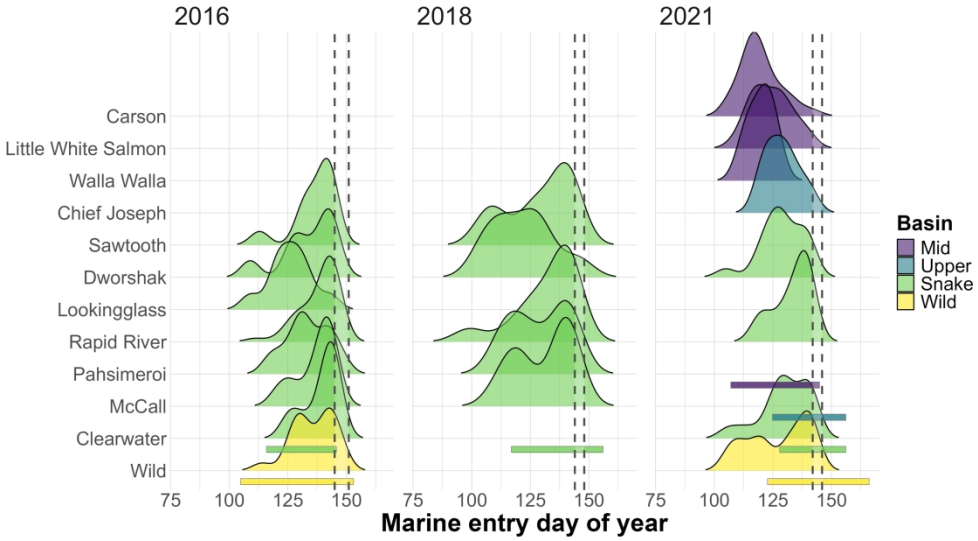


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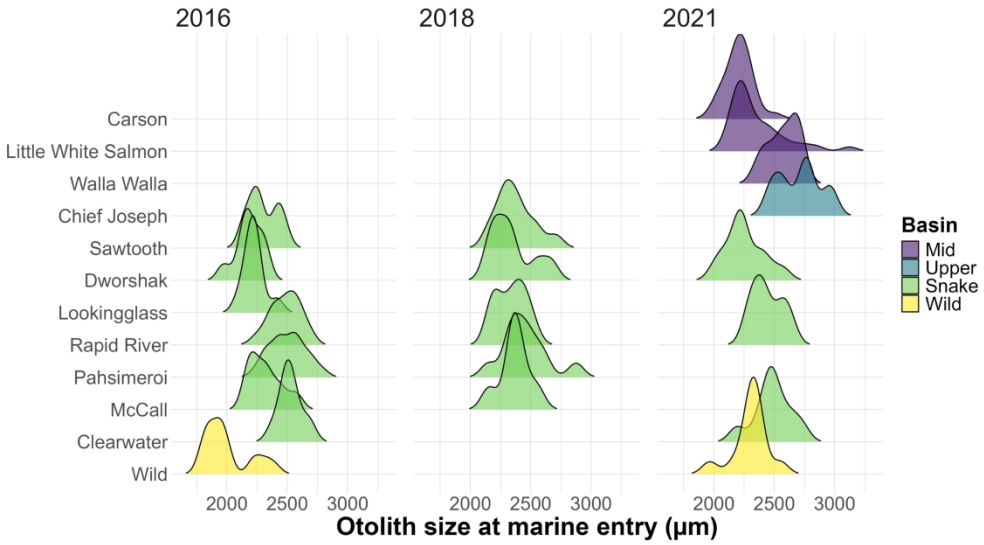


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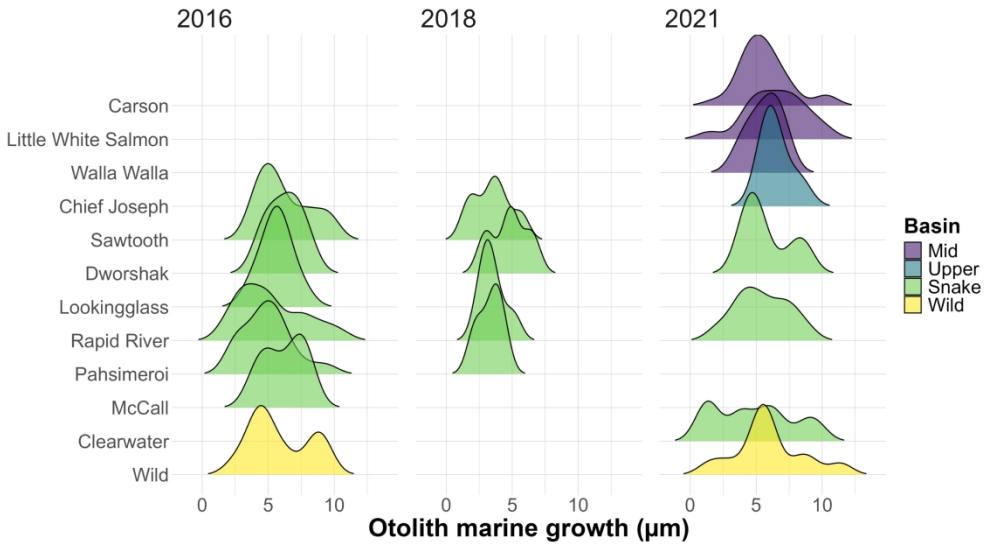


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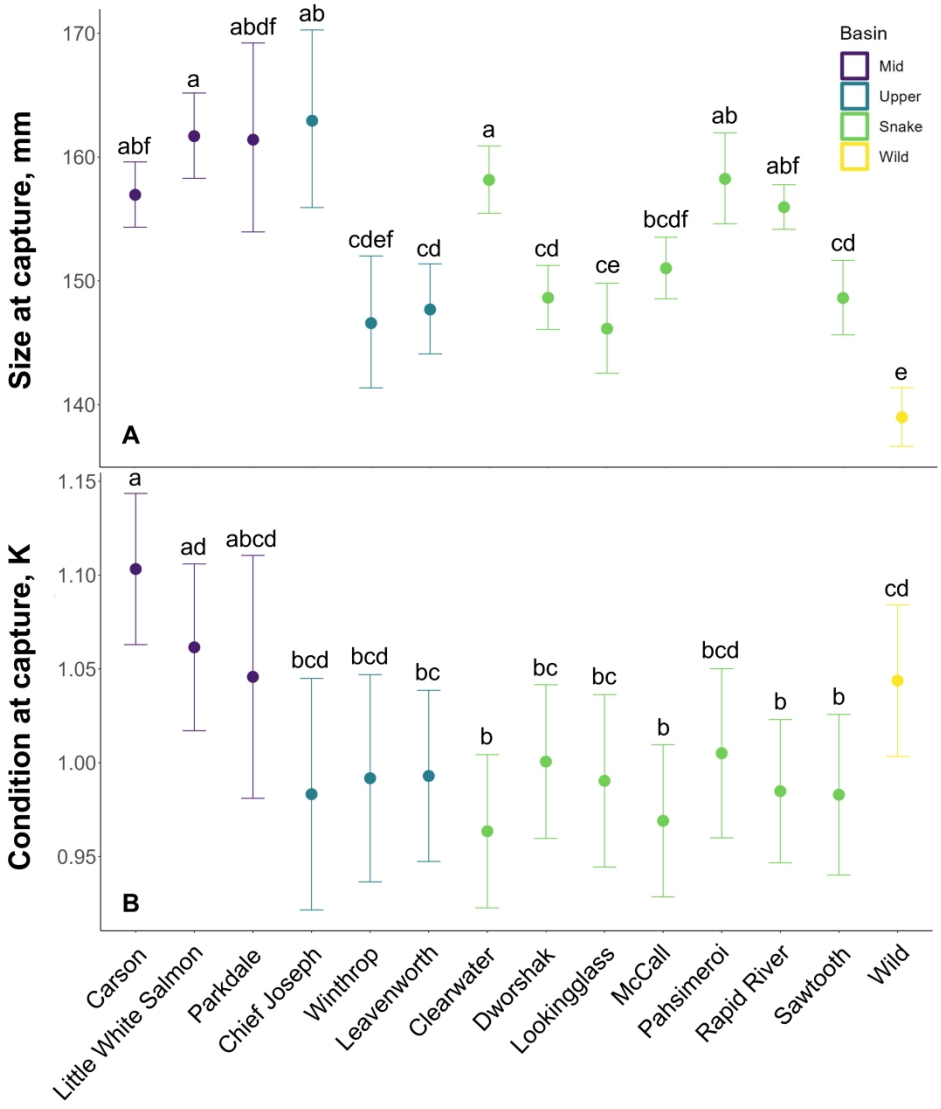


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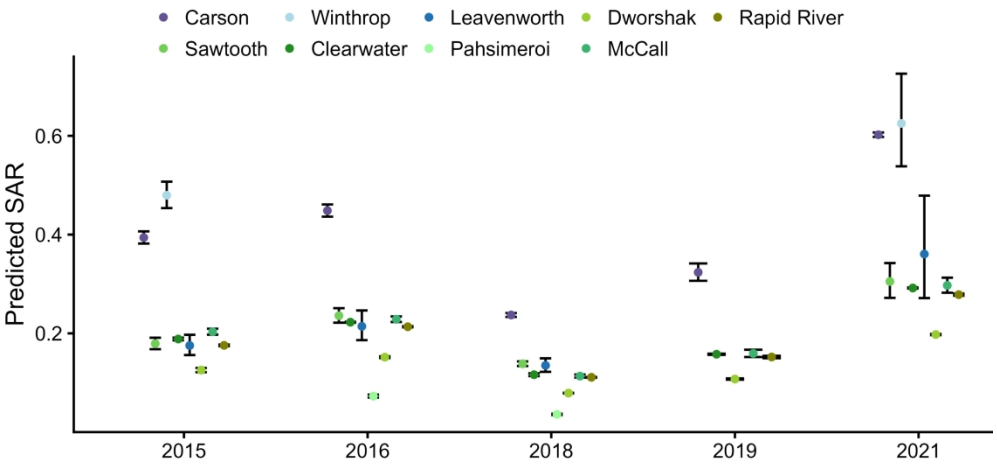


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